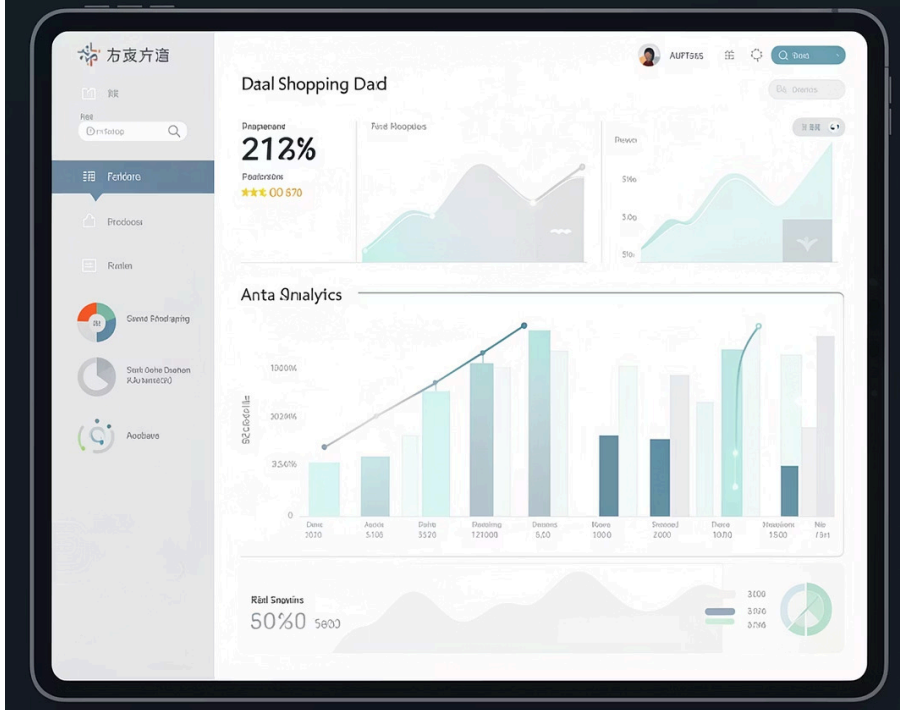
Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions**



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Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating only

50

Locations

Demographics, purchase details, and shopping behavior across 4 product categories

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Data Preparation in Python

→ Load & Explore

pandas, df.info(), .describe()

→ Clean

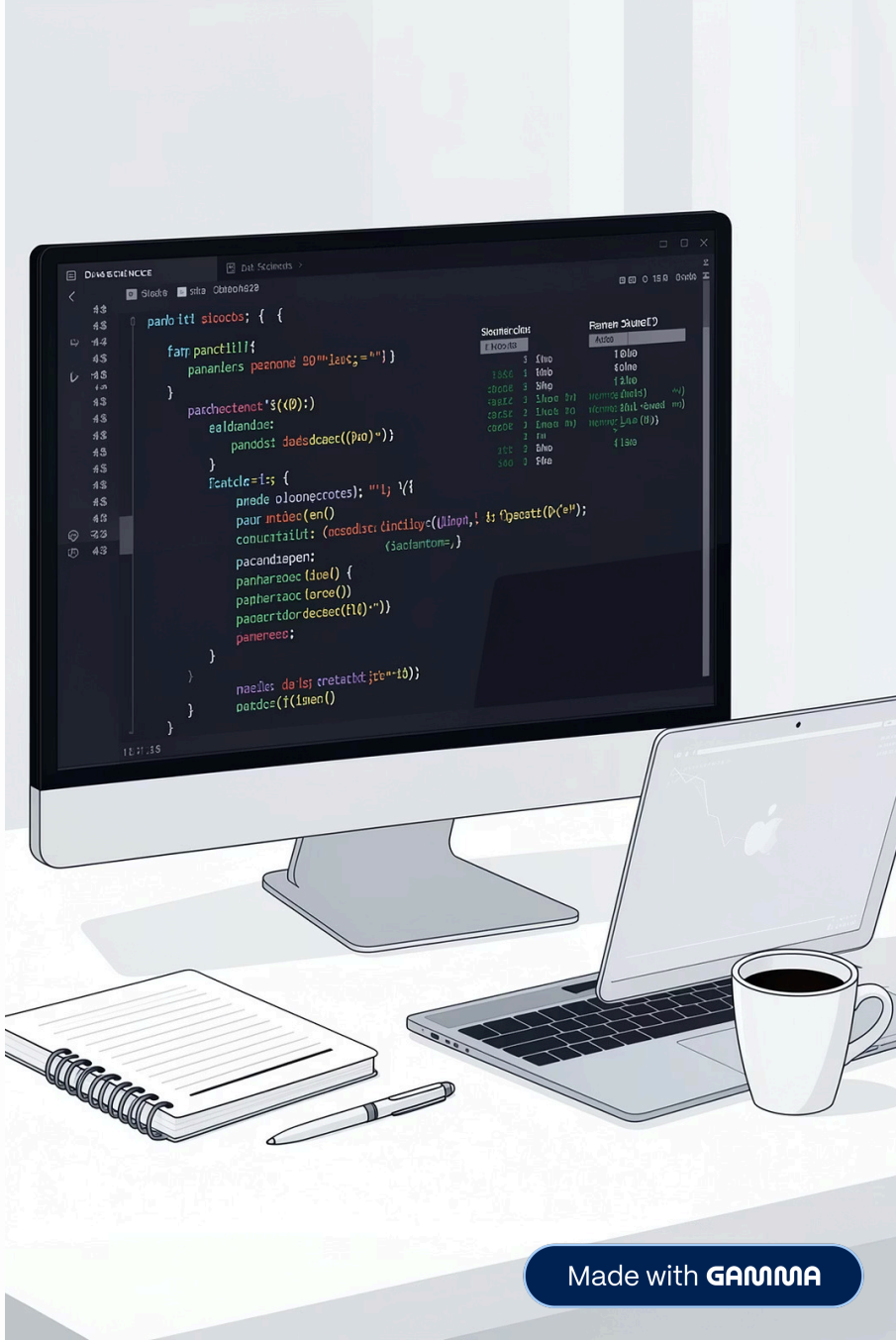
Imputed missing Review Ratings by category median

→ Engineer

Created age_group and purchase_frequency_days

→ Load to DB

Connected to PostgreSQL for SQL analysis



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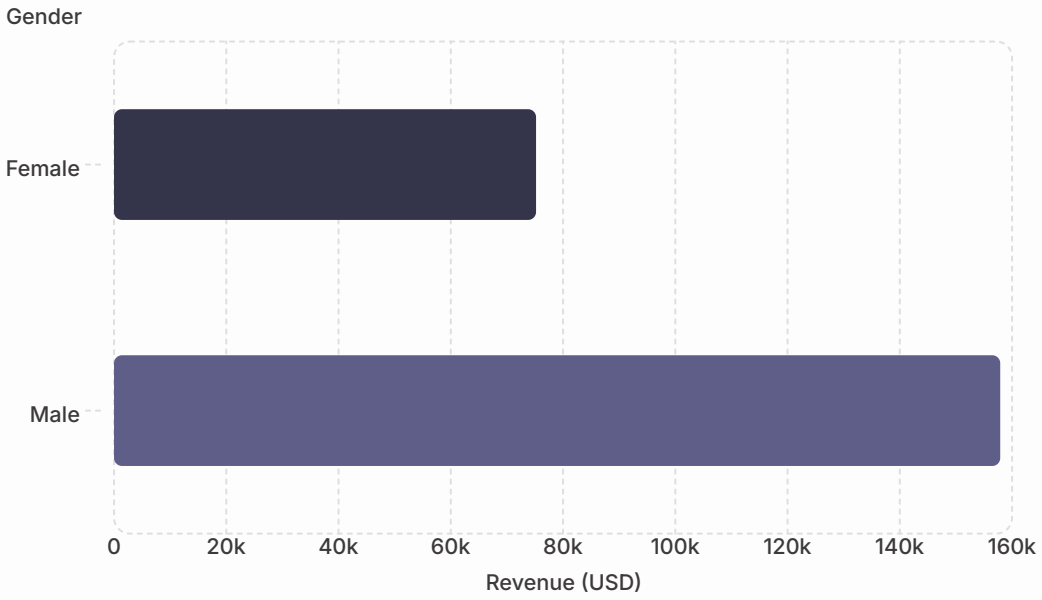
Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers

- Male: $157,890
- Female: $75,191

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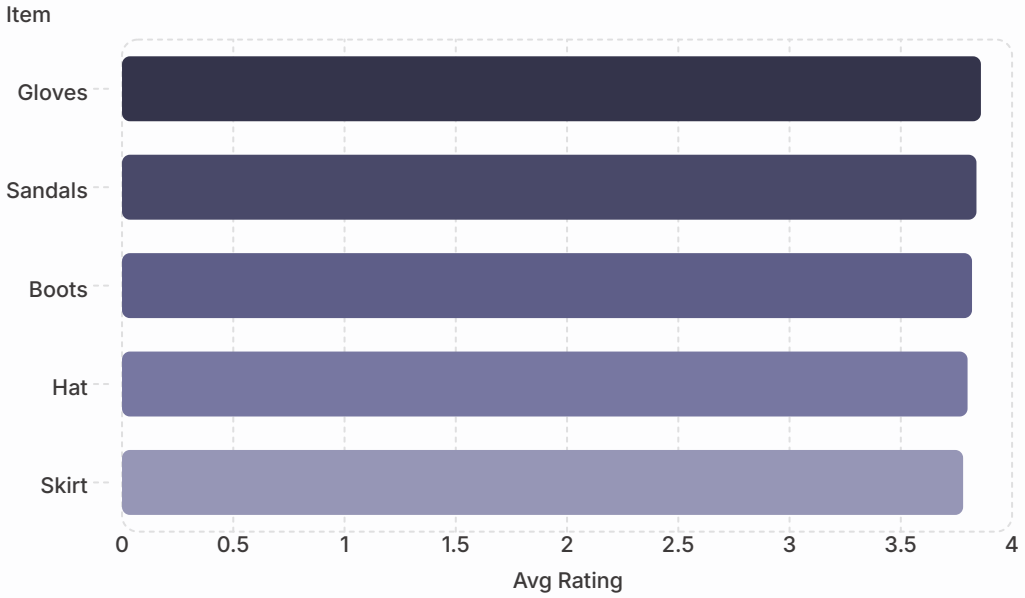
Top Products & Shipping Insights

Top 5 Products by Rating



Shipping Type Comparison

- **Express:** $60.48 avg purchase
- **Standard:** $58.46 avg purchase

Express buyers spend slightly more on average

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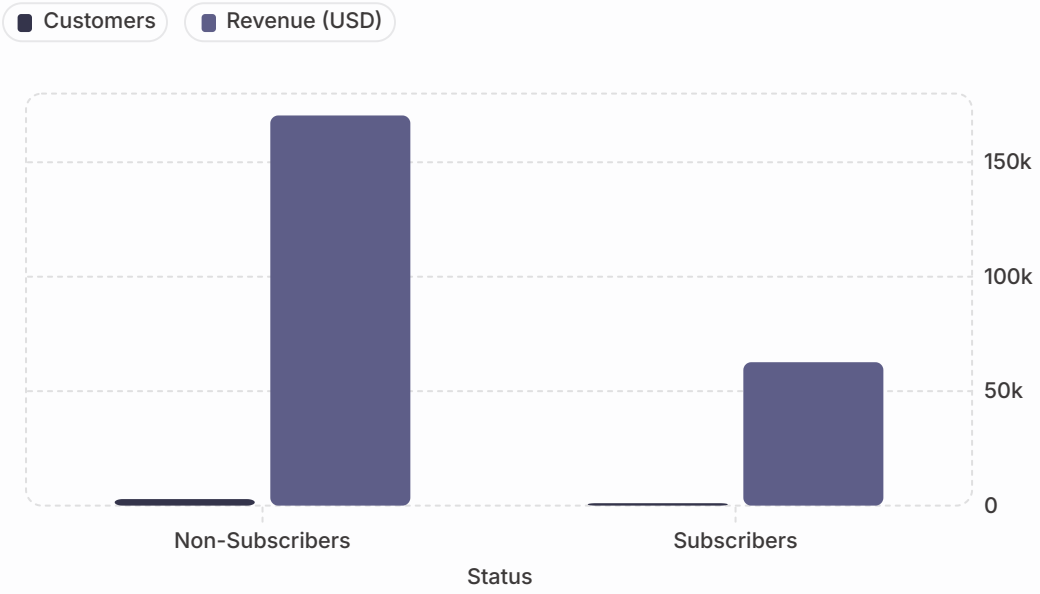
Subscribers vs. Non-Subscribers



Subscription Breakdown

73% Non-Subscribers

2,847 customers · $59.87 avg spend

27% Subscribers

1,053 customers · $59.49 avg spend

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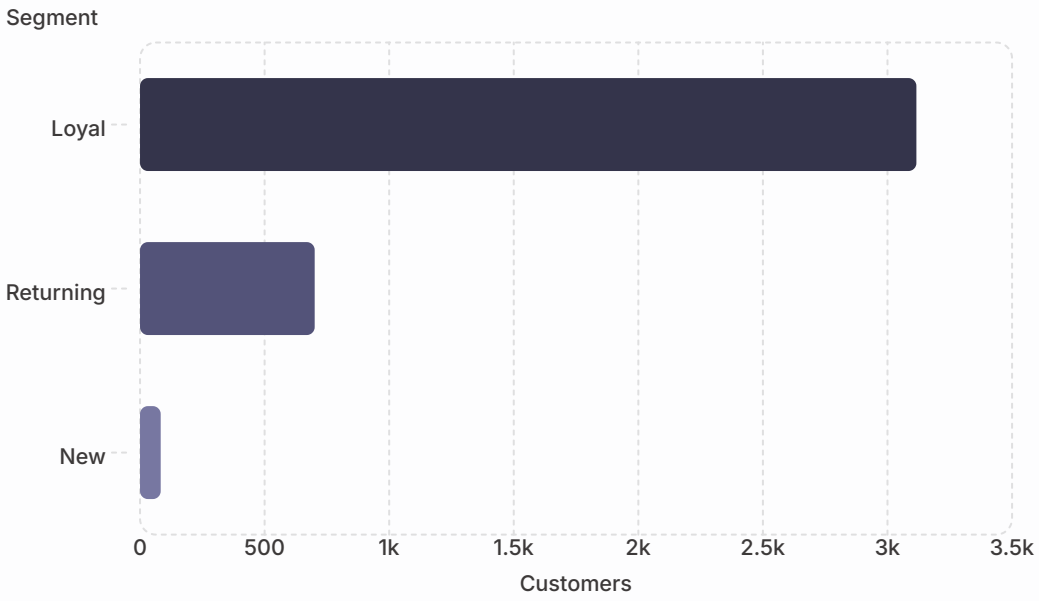
Customer Segmentation



Segment Definitions

Loyal

3,116 customers

Returning

701 customers

New

83 customers

839 customers used discounts while spending above average

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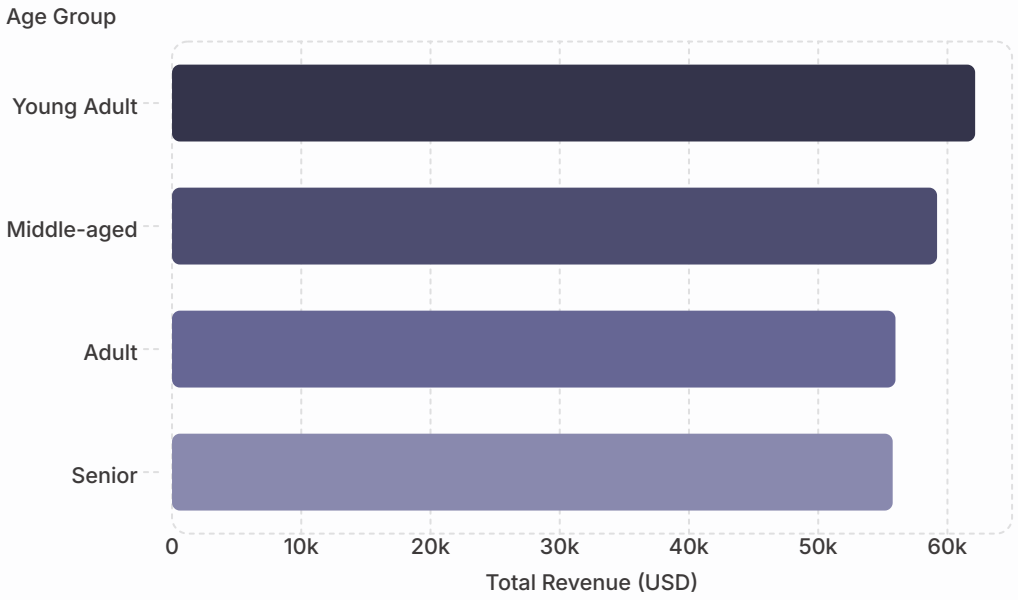
Revenue by Age Group



Age Distribution

Average customer age: **44 years**

Age range: 18–70

- Young Adults lead in revenue
- Revenue fairly balanced across groups

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Power BI Dashboard

3.9K
Customers

$59.76
Avg Purchase

3.75
Avg Rating

Filters: Subscription Status, Gender, Category, Shipping Type

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discounts

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups & express users

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